

TDM 729.89 915.51 185.62▲25.43% FLR 660.27 745.28 85.01▲12.88%  
HUM 749.73 924.29 174.56▲23.28% UVD 155.59 181.57 25.98▲16.70%  
DMW 833.72 1004.01 170.29▲20.43% QUV 440.55 540.21 99.66▲22.62%  
YZJ 903.49 1127.46 223.97▲24.79% HZT 265.51 344.98 59.47▲20.83%  
GLY 982.07 1219.39 237.32▲24.17% PCW 811.44 1029.66 218.22▲26.89%  
VDA 113.74 143.41 29.67▲26.09% AIK 361.77 451.39 89.62▲24.77%  
UVV 468.08 535.41 67.33▲14.38% ZJJ 858.36 994.57 136.21▲15.87%  
HLE 565.49 670.03 104.54▲18.67% ZHJ 894.39 1046.68 151.89▲16.97%  
HLE 565.49 670.03 104.54▲18.67% ZHJ 894.39 1046.68 151.89▲16.97%

MOULIK JAIN | DS 402

PPJ 912.63 1038.36 125.73▲13.78% ZGW 391.59 491.48 99.89▲25.51%  
UAD 1309.55 1655.62 346.07▲26.43% BNY 969.21 1130.65 161.44▲16.66%  
DAO 1295.17 1641.66 346.49▲26.75% SDM 735.44 913.39 177.95▲24.20%  
PNR 654.33 775.84 121.51▲18.57% TQU 1323.91 1646.42 322.51▲24.36%  
FTM 1112.21 1372.21 260.00▲23.37% OIS 543.42 667.24 123.82▲22.79%  
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# Graph-Based Financial Intelligence: Detecting Money Laundering with GNNs

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# Motivation

\$2 Trillion Laundered Annually, Detection is Broken

- Traditional rule-based systems produce 95%+ false positive rates, analysts are overwhelmed
- Each transaction looks normal in isolation; **the fraud is in the pattern across accounts**

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## Two Real-World Benchmarks

### IBM AML HI-Medium

- 2M transactions, ~500k accounts
- 0.003% fraud rate
- 8 laundering pattern types (fan-out, cycle, stack...)
- Edge-level classification

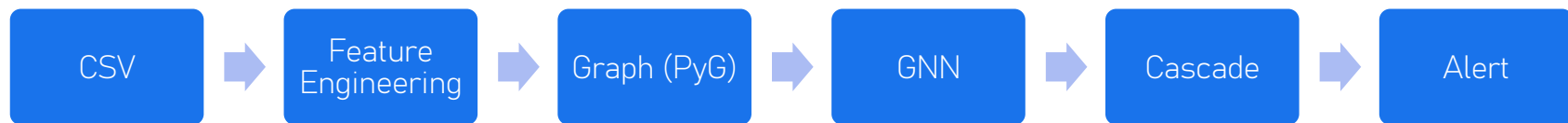
### Elliptic Bitcoin

- 203k nodes, 234k edges
- 9.76% fraud rate
- Cross-domain generalization test

60/20/20 stratified split · Focal Loss for class imbalance · V100 GPU on PSC Bridges2

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## Pipeline: Raw Transactions → Alert

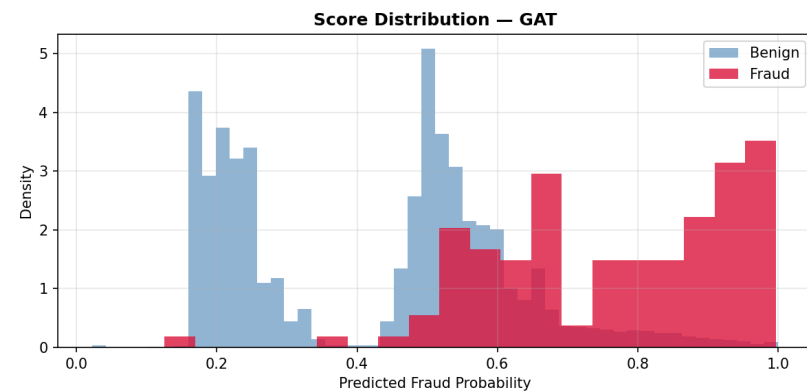


*Features:* 16 node feats (PageRank, k-core, degree stats) + 11 edge feats  
(structuring flags, delta-t, amount)

*Models tested:* GraphSAGE · GAT · TGN · H2GCN

# Why Standard GNNs Struggle and How I Fixed It

- Fraudsters deliberately route money through legitimate accounts
- Standard message passing **averages** neighbors → fraud signal gets washed out
- H2GCN fix: keep ego-embedding separate; concatenate [self + 1-hop + 2-hop] instead of averaging



GAT assigns high fraud scores to tens of thousands of benign transactions, precision collapses at any threshold

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## Graph Structure Beats Tabular — H2GCN Leads

| Model              | Recall       | ROC-AUC      | PR-AUC       |
|--------------------|--------------|--------------|--------------|
| RF (no graph)      | 0.718        | 0.977        | 0.152        |
| XGBoost (no graph) | 0.581        | 0.972        | 0.511        |
| GraphSAGE          | 0.798        | 0.947        | 0.042        |
| GAT                | 0.960        | 0.902        | 0.005        |
| TGN                | 0.623        | 0.974        | 0.136        |
| H2GCN              | <b>0.903</b> | <b>0.960</b> | <b>0.113</b> |

*H2GCN achieves the best ROC-AUC of any GNN and 22× better PR-AUC than GAT*

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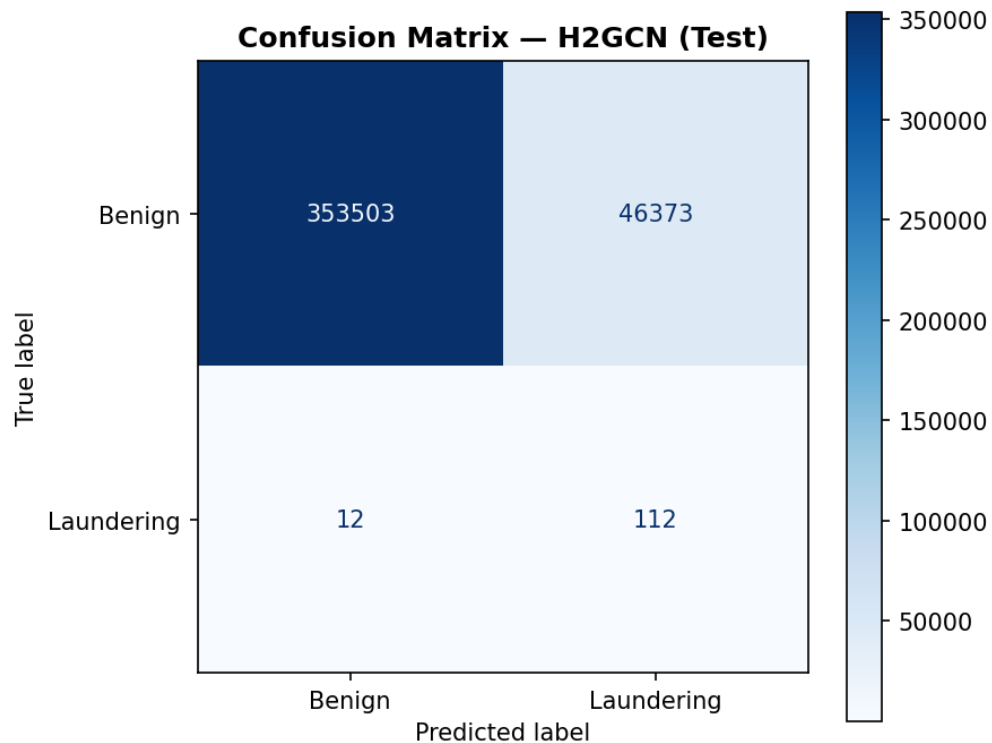
# High Recall Alone Is Not Deployable

The problem (one stark stat):

- GAT flags 96% of fraud.
- Precision = **0.06%**.
- Every real fraud alert comes with **1,667 false alarms**.

The solution table:

|           | H2GCN | H2GCN + Cascade |
|-----------|-------|-----------------|
| Recall    | 90.3% | 54.8%           |
| Precision | 0.24% | 82.9%           |
| F1        | 0.005 | 0.660           |
| PR-AUC    | 0.113 | 0.595           |
| ROC-AUC   | 0.960 | 0.993           |

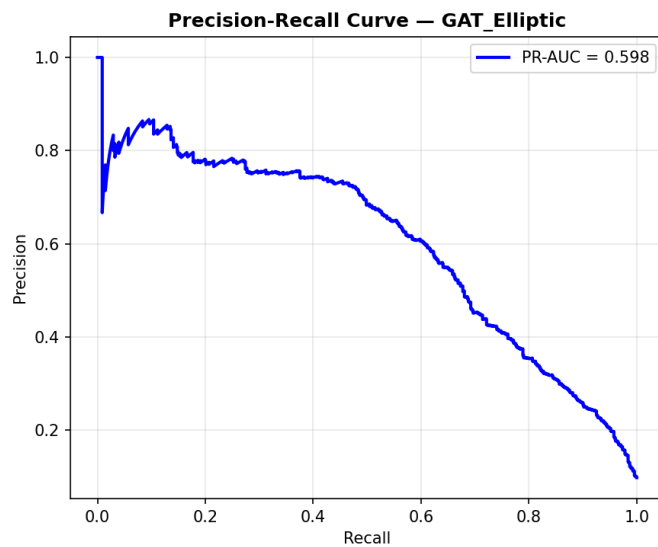


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46,373 false positives, still a problem the cascade solves



# Generalises to Elliptic Bitcoin



GAT on Elliptic: 94.4% recall · PR-AUC = 0.599 · ROC-AUC = 0.906



Different domain, different asset class, different fraud rate; the GNN approach transfers

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# Conclusion & Future Work

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## Key Takeaways

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Graph structure is necessary — GNNs consistently beat tabular baselines on recall

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Heterophily is the core challenge — H2GCN's ego-separation gives 22× better PR-AUC than GAT

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The two-stage cascade (GNN → XGBoost) is the right production design: F1 = 0.66, precision = 82.9%

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## Future work

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Scale to full 13M-edge dataset with heterogeneous bank-account graph

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Deploy TGN memory module for real-time streaming detection

**Citations:** Weber et al. 2019 · Zhu et al. 2020 (H2GCN) · Hamilton et al. 2017 (GraphSAGE) · Lin et al. 2017 (Focal Loss) · Rossi et al. 2020 (TGN)



*Graph structure  
is not just useful  
for AML, it is  
essential.*

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THANKS!